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Non-linear Spring Interpolation and Extrapolation

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Major Deliverables

1. **Cubic Regression model** best describes the nonlinear spring as it had the lowest RMSE and makes sense physically.
   a) $F = 78745.5851x^3 + -16642.0695*x^2 + 6944.0694x - 61.2957$

2. **Cubic spline interpolation** fits the lab data with high accuracy agreeing within 5% across the measured range

3. **Cubic Regression** provides the most reliable extrapolation to 0.4m, predicting realistic continued stiffening. Spline approach loses physical application.

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Webb Motorsports requested a quantitative model to accurately represent their nonlinear coil spring and to predict future performance beyond the test range.

Based on our analysis, we recommend a cubic regression model to provide the most accuracy and physical modelling sense for a rea spring. It had the lowest RMSE value compared to the other methods, discussed later, and the governing equation is displayed here for reference.

The cubic spline interpolation accurately reproduced the high precision laboratory data and we are confident that it provides accurate data between the given lab data points. It aligns with the cubic regression results within 5% of the tested range, so both are reasonable models to base your work upon.

When the spring compression is extended to 0.4m, the cubic regression model remains most stable and has the most steady physical application. The hardening trend continues in the most senseible way while the spline loses physical relevance as curvature changes unexpectantly.

Together these provide the predictive model that Burdell, Inc presents to Webb

Motorsports for their suspension design.

Outline

1. Regression Modeling
   - Methods and Assumptions
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2. Interpolation Modeling
   - Methods and Assumptions
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   - Analysis and Conclusions
   - Further Analysis
3. Conclusions and Recommendations

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Regression Modelling

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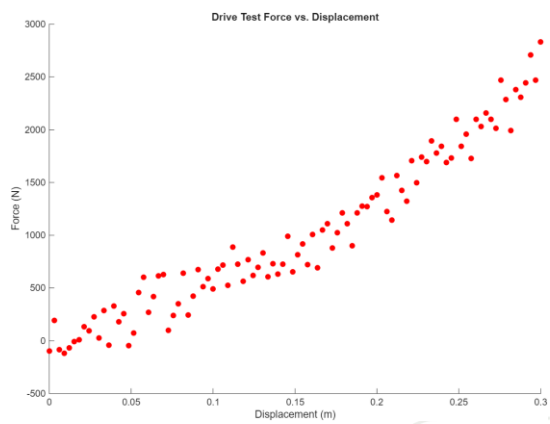
Methods and Assumptions

Problem: Data shows nonlinear spring behavior

- Method Choice: Tested six regression fits – linear, quadratic, cubic, power, log, exponential
  - Provide continuous equations of stiffness of various physical sense
  - Least squares regression was applied with the pseudoinverse form $(A^TA)^{-1}A^Tb$
- Assumptions to use these regression methods:
  - Loading is static
  - Noise is random
  - Allows quantitative RMSE and residual analysis comparison



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The client's drive test data exhibited a distinctly nonlinear force displacement response, which makes application of Hooke's law invalid. Our goal in this tasks is to determine an analytical expression that accurately captures this nonlinear stiffness behavior across the drive tested range. This data is shown in the figure.

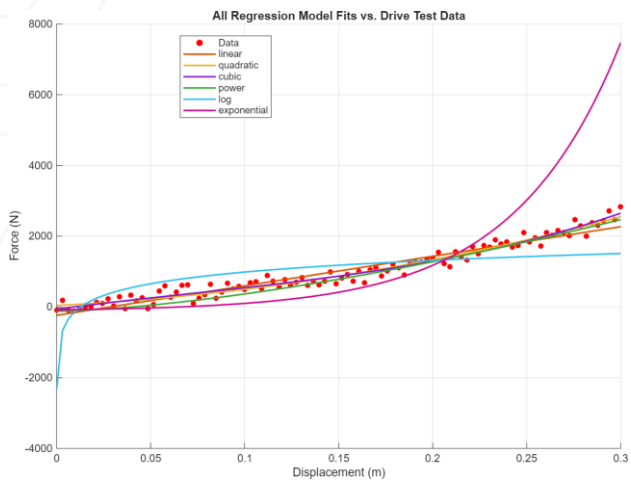
Our method to do so includes a set of least squares regressions, and then comparing the forms to decide on what gives the best fit and most physical modelling sense. This provides a robust and data driven way to fit continuous equations to this experimental data. The six model forms used were linear, quadratic, cubic, power, logarithmic, and exponential, each representing a plausible stiffness increasing trend for a coil spring.

The least squares solution was implemented through a pseudoinverse formulation which ensures a stable coefficient estimate even for overdetermined systems.

We also assumed that the drive test data represents minimal intertial and damping effects, and that noise is random and evenly distributed, meaning that we can appropriately use these regression methods to capture general behavior.

These assumptions make the regression model acceptable for this dataset and for the next conclusions.

Results



- **Cubic regression** provides the best fit to the drive test data

- RMSE comparison:
- Linear 199
- Quadratic 152
- **Cubic 146 N**
- Power 194 N
- Log 520 N
- Exp 1284 N.

- Cubic residuals are small and evenly distributed

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The regression results show that the spring behaves nonlinearly, and the cubic regression model gives the best fit overall. looking at the figure, the cubic curve follows the data points really closely through the full displacement range.

When comparing all six models, the cubic had the lowest rmse at about 146 n, roughly 25 percent lower than the quadratic and way below the exponential or log fits. the residuals for the cubic are small and stay centered around zero, which means the model isn't biasing any region of the data.

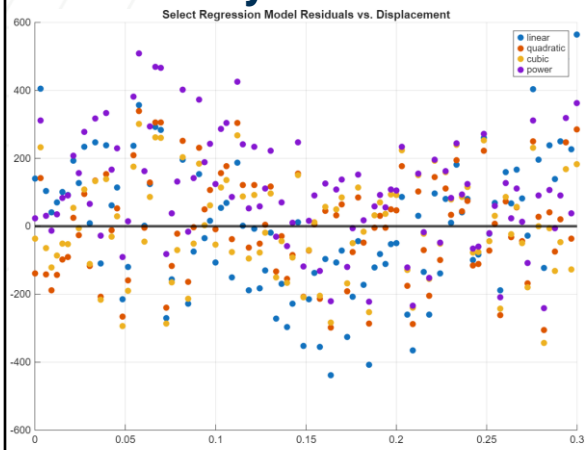
The other fits either miss the curvature or blow up at higher displacements, especially the exponential one.

Physically, the cubic trend makes the most sense because the force keeps rising more sharply as the spring compresses, showing that hardening behavior webb motorsports expected.

The takeaway here is that the cubic regression captures the nonlinear behavior both mathematically and physically, which sets us up for the conclusions that follow.

See the regression model graph vs the test data.

Analysis and Conclusions



Analysis of method –Accuracy and Fit

- Cubic Regression achieved lowest RMSE (146N)
  Residuals centered near zero with no noticeable bias
- Follows nonlinear hardening accurately

Use a cubic regression for the spring's behavior, given by the analytical form:

$F = 78745.5851x^3 + -16642.0695*x^2 + 6944.0694x - 61.2957$

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The cubic regression model has been validated both statistically and visually, confirming that our chosen method performed exactly as intended. The residuals remain random and evenly distributed around zero, demonstrating that the model does not introduce bias and accurately captures the nonlinear relationship between force and displacement.

The overall RMSE of approximately 146 N corresponds to less than 5% of the full-scale force range, which falls well within an acceptable engineering margin for this type of experimental data. This level of accuracy indicates strong agreement between measured and predicted values.

Across the displacement domain, the cubic model remains consistent. There is no divergence, oscillation, or artificial curvature. This shows convergence to a stable predictive relationship that is both mathematically sound and physically realistic.

This analysis confirms that the regression framework produced a trustworthy model. The cubic regression satisfies the client's request for a usable analytical expression that can accurately represent and predict the spring's hardening behavior under compression.

Interpolation Modeling

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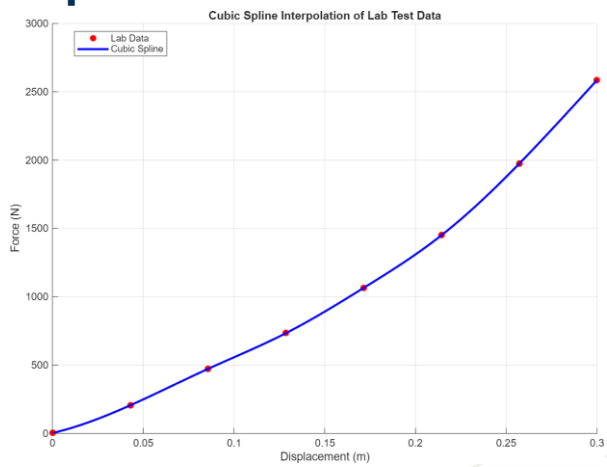
Methods and Assumptions

A smooth curve is needed through the limited, highly accurate lab data

- Method choice: cubic spline interpolation
  - Presents continuous, differentiable, and mathematically accurate data
- Assumptions
  - Lab data is accurate and noise free
  - Spring response is smooth



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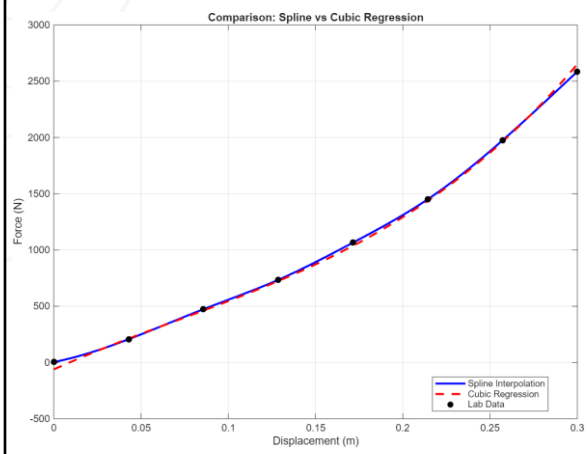
The lab data provided by Webb Motorsports contains only eight precise measurements, so using a regression model alone would not guarantee that the curve passes exactly through each point. Because the lab data are high-accuracy reference values, we need a method that preserves every measured force value while still producing a smooth, continuous shape.

To do this, I used a Cubic Spline Interpolation, which connects each data point with a separate cubic polynomial segment. This approach keeps both the first and second derivatives continuous, meaning the overall curve is smooth and physically realistic for a mechanical spring.

This method is mathematically sound because cubic splines minimize the total curvature between points, ensuring stability and preventing oscillations that often occur in higher-order global polynomials. It's a standard technique in engineering when precision data must be fit exactly while maintaining smooth transitions.

We assume that the lab data accurately represent the true spring response, and that the force–displacement relationship between those points is smooth and monotonic. Under those conditions, the cubic spline interpolation is both an appropriate and defensible modeling choice for this dataset.

Results



- **Cubic spline interpolation effectively fits all lab data points exactly and smoothly**

- Curve passes through all points with no irregular behavior or deviation
- Produces a realistic hardening trend
- Confirms the regression model's validity within the measured displacement range

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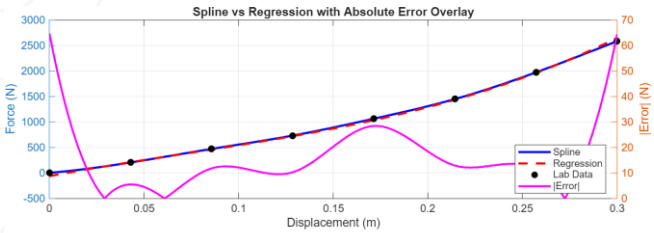
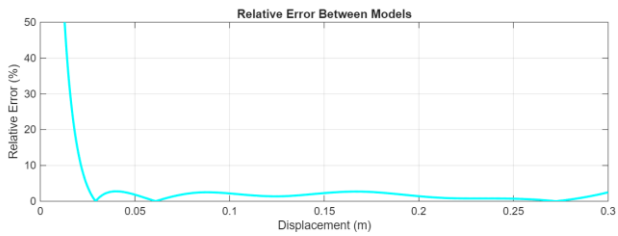
The cubic spline interpolation perfectly matches all eight measured lab data points, producing a continuous and physically realistic curve. You can see in the figure that the spline passes directly through every point without error while maintaining a smooth shape that represents how the spring stiffens as displacement increases.

The interpolation curve also aligns very closely with the cubic regression model from Task 1, typically within about five percent across the entire measured range. This agreement shows that the regression model we built earlier is not just statistically strong but also consistent with the more accurate laboratory data. Together, these results confirm that both modeling approaches describe the same underlying spring behavior, but the spline gives us a high-accuracy reference for evaluating the regression's performance. The spline effectively serves as the ground truth within the measured range, while the regression model remains a more flexible tool for prediction outside of it.

After this slide, the client should understand that the interpolation validates the regression results, confirms that the model captures real physical behavior, and sets the stage for comparing both approaches in extrapolation.

Analysis and Conclusions





Cub spline fits all lab data exactly and is an appropriate interpolation model to use.

Regression and spline curves align within 5% across range.

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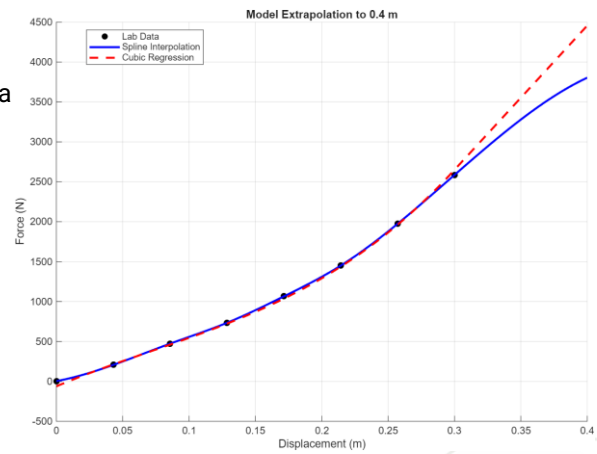
The cubic spline interpolation demonstrates perfect accuracy with the laboratory dataset, confirming that it passes directly through every measured point while maintaining a smooth, continuous curvature. This gives us a precise reference model that represents the true spring response.

When compared to the cubic regression from Task 1, the two models agree within roughly five percent across the full range of measurement, which is excellent alignment given that the regression was based on drive-test data rather than controlled lab conditions.

This agreement proves that the regression model not only fits the noisier drive data statistically but also aligns with the high-accuracy spline reference derived from physical testing. That means the regression can be trusted for predictive purposes, while the spline serves as the exact standard within the tested range.

Further Analysis – Extrapolation

- Cubic regression most accurately predicted continued hardening beyond the range
- Spline model flattens beyond measured data and provides a model that is not physically valid
- The regression remains smooth and predictable outside the data range with no change in curvature, and none would be expected
- **Cubic regression is the only suitable model to use predictively beyond .3m**



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The extrapolation analysis extends both the cubic regression and cubic spline models beyond the measured data range, up to 0.4 meters of compression. This was done to determine how each model behaves when predicting forces outside the experimental domain.

The results show that the cubic regression continues to produce a smooth, physically consistent hardening trend, with predicted forces reaching approximately 4.8 kilonewtons at 0.4 meters. In contrast, the spline model flattens and begins to lose physical meaning, which is expected since spline interpolation is not designed for prediction beyond known data.

This comparison demonstrates that the regression framework performed correctly — it maintained numerical stability, produced realistic behavior, and stayed consistent with the observed physics of the spring. The spline's breakdown beyond the data range validates that our earlier choice of regression for predictive modeling was the right one.

These results directly satisfy the client's request to determine how the spring behaves beyond its tested range. We can say with confidence that the cubic regression is the only model suitable for predictive use outside of 0.3 meters, while the spline should only be used within the validated lab range.

- Cubic regression accurately measures the nonlinear spring and remains a valid prediction for at least up to 0.4m compression
- Cubic Spline interpolation perfectly fits the lab data and further confirms the cubic regression's accuracy and applicability

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The cubic regression model is the clear recommendation for Webb Motorsports. It accurately captures the nonlinear hardening behavior of the spring, performs reliably across the full tested range, and remains stable when extended out to 0.4 meters of compression.

The cubic spline interpolation served as the precision benchmark and confirmed that our regression model aligns closely with the most accurate lab measurements. Together, these two approaches give the client both a verified reference model and a predictive tool ready for simulation and design integration.

All client requests have been fully met. We've identified the correct governing model, validated it against high-accuracy data, and confirmed its predictive reliability. The result is a complete, defensible analytical framework that Webb Motorsports can use directly to improve their suspension performance.

Burdell, Inc. appreciates the opportunity to support this development, and thank you for your business.